CERTIFICATE

This is to certify that the project entitled "**Real Time Sign Language Detection**", submitted **Sahil Siddique** (Enrollment No. 2200100808), **Shahbaz Husain Khan** (Enrollment No. 2200101606), **Tanzeer Reyaz** (Enrollment No. 2200101516), in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering at Integral University, Lucknow (India), is a record of the students' own work carried out under the supervision and guidance of .

The project report embodies the results of original work and studies conducted by the students and have not formed the basis for the award of any other degree or diploma to the candidates or to anyone else.

Mr. Danish Ahmad Assistant Professor (Project Guide)	Dr.Shish Ahamad Professor and Head Department of CSE

DECLARATION

I/We hereby declare that the project entitled **"Real Time Sign Language Detection"** submitted by me/us in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering, of Integral University, Lucknow, is record of my/our own work carried under the supervision and guidance of **Mr. Danish Ahmad** (Assistant Professor)

To the best of my/our knowledge this project has not been submitted to Integral University, Lucknow or any other University or Institute for the award of any degree.

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We would like to express our sincere gratitude to all those who supported us throughout the course of our final year project, "**Real Time Sign Language Detection**".

First and foremost, we extend our heartfelt thanks to our project supervisor, Mr. Danish Ahmad, for his constant guidance, valuable insights, and unwavering support throughout the project, His encouragement and constructive feedback played a vital role in shaping our work.

We are also grateful to the faculty and staff of the Department of Computer Science and Engineering, for providing us with the knowledge, infrastructure, and technical assistance necessary for the successful completion of our project.

We would like to acknowledge the cooperation and team spirit of our group members **Sahil Siddique**, **Shahbaz Husain Khan** and **Tanzeer Reyaz** whose dedication and collaborative efforts were instrumental in achieving our objectives.

Lastly, we are thankful to our families and friends for their moral support and encouragement throughout this academic journey.

Sahil Siddique **Shahbaz Husain Khan** **Tanzeer Reyaz**

PREFACE

It gives me immense pleasure to present this project report on **"Real Time Sign Language Detection"**, which was undertaken as a part of our final year curriculum. The inspiration behind choosing this topic stems from the growing need for effective communication solutions for people with hearing and speech impairments, and the increasing role of artificial intelligence in solving real-world problems.

This project focuses on developing a real-time system that can detect and interpret sign language gestures using computer vision and machine learning techniques. Through this work, we aimed to bridge the communication gap by enabling seamless interaction between sign language users and non-sign language users.

I would like to express my heartfelt gratitude to **Mr. Danish Ahmad**, our project guide, for her continuous guidance, valuable suggestions, and constant encouragement throughout the course of this project. I am also thankful to our department faculty members and my teammates **Sahil Siddique**, **Shahbaz Husain Khan** and **Tanzeer Reyaz** for their contributions and collaboration.

Working on this project has significantly enhanced my understanding of machine learning, computer vision, and real-time system development. It has also provided valuable insights into how AI can be used to create inclusive technologies that benefit society. I hope this report provides useful knowledge to readers interested in assistive technologies and intelligent systems.

Sahil Siddique **Shahbaz Husain Khan** **Tanzeer Reyaz**

ABSTRACT

Real-time sign language detection systems play a crucial role in bridging communication gaps between hearing-impaired individuals and the general population. This project presents a deep learning–based real-time sign language recognition system that interprets hand gestures and converts them into meaningful text output. The system is designed to provide an intuitive and accessible interface for seamless human-computer interaction.

The proposed system leverages computer vision techniques and machine learning models to detect and classify hand gestures from live video streams. A Convolutional Neural Network (CNN) model is trained on labeled sign language datasets to extract spatial features from hand gestures, while real-time video processing is implemented using OpenCV. The trained model is capable of recognizing multiple hand signs with high accuracy and low latency.

The application is integrated with a frontend built using React Native, enabling cross-platform mobile support. Communication between the frontend and backend is handled via APIs or socket-based connections to ensure real-time predictions. The backend processes incoming frames, performs gesture classification, and returns the predicted text instantly.

To improve system performance, preprocessing techniques such as background subtraction, image normalization, and hand region segmentation are applied. The system also addresses challenges such as varying lighting conditions and different hand orientations.

Experimental results demonstrate that the system achieves high accuracy in real-time gesture recognition while maintaining efficient performance suitable for deployment on mobile devices. This project highlights the practical implementation of AI-driven assistive technology and has potential applications in education, accessibility tools, and inclusive communication platforms.

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LIST OF ABBREVIATIONS AND SYMBOLS USED

Abbreviation / Symbol	Full Form / Meaning
CSV	Comma-Separated Values
CNN	Convolutional Neural Network
CV	Computer Vision
CPU	Central Processing Unit
GPU	Graphics Processing Unit
GFD	Data Flow Diagram
ER	Entity Relationship
FPS	Frames Per Second
GB	Gigabyte
ML	Machine Learning
NLP	Natural Language Processing
OpenCV	Open Source Computer Vision
OS	Operating System
RAM	Random Access Memory
RST	Representational State Transfer
ROI	Region of Interest
UI	User Interface
UX	User Experience
URL	Uniform Resource Locator
AI	Artificial Intelligence

Abbreviation / Symbol	**Full Form / Meaning**
RGB	Red Green Blue
DFD	Data Flow Diagram
DL	Deep Learning
FPS	Frames Per Second
ID	Identifier

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